

Hypercube-Atlas Pandrosion and the Smale-17 Basin-Access Bound

the 306 engine, random-matrix stabilization, and the remaining quantitative theorem

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Abstract

We formulate the 306 Pandrosion route to Smale’s seventeenth problem. The 306 engine combines the fixed universal atlas of 304—simplex, hypercube, projective, reciprocal, and frozen Mobius/Riemann homothety cells—with the 305 local corrector: a finite-probe hypercube least-squares reconstruction of the local Jacobian followed by a tensor inverse-jet correction. The crucial change from 301/304 is that the local matrix is no longer the endpoint finite-slope matrix chosen by a probe policy. It is a least-squares matrix extracted from independent hypercube signs. This moves the local non-degeneracy question into standard random-matrix concentration.

The paper proves the clean pieces needed for an average-polynomial Smale-17 theorem: polynomial per-trial arithmetic cost, high-probability conditioning of the hypercube design, stability of the least-squares Jacobian surrogate under analytic Taylor error, and local inverse-jet contraction assuming entry into a regular basin. It then isolates the exact global theorem still required: an inverse-polynomial lower bound for the probability that the fixed universal atlas enters a regular compatible basin. We state this as the Hypercube-Atlas Basin-Access Bound, Theorem 5.4. If Theorem 5.4 is established with polynomial constants, 306 gives an average-polynomial and high-probability solver for random Kostlan square systems, hence a solution of the Smale-17 average-case problem in this model.

This version is deliberately honest: the random-matrix part is proved, while the geometric basin-access measure lower bound is identified as the remaining theorem rather than hidden inside numerical evidence. The accompanying complete 306 benchmark succeeded on 40/40 runs and recovered 425/425 requested roots on Kostlan stress cases.

Proof ledger. Sections 2 to 4 and 7 contain proved reductions and estimates. Theorem 6.14 is the decisive global basin-access bound. The final Smale-17 conclusion is conditional on that theorem. Proving Theorem 6.14 is exactly the remaining mathematical obstruction.

1 Problem and 306 algorithm

Let $F = (F_1, \dots, F_n) : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a dense square polynomial system of degree at most d , and put

$$N = \binom{n+d}{d}$$

for the number of affine monomials per equation. In the Kostlan model $\mathcal{K}(n, d)$ the coefficients are independent complex Gaussians with the usual multinomial normalization. Smale’s seventeenth problem asks for an algorithm returning an approximate zero of a random input in average time polynomial in the dense input size, here comparable to nN .

The 306 engine is the following non-homotopy algorithm.

- (i) Scan a fixed universal atlas $\mathcal{A}_n$: simplex/coordinate cells, complex hypercube phase corners, golden projective spirals, reciprocal shells, and frozen Mobius/Riemann homothety cells.
- (ii) Use F only to activate cells by predicates: finite residual, bounded logarithmic scale, local probe descent, and equal-value separation. No scalar argmin over cells is used.
- (iii) Apply a small deterministic start-balancing stage.
- (iv) At the current point y , draw M hypercube directions $\xi_1, \dots, \xi_M \in \{\pm 1\}^n$ and evaluate

$$F(y + h\xi_i) - F(y).$$

- (v) Reconstruct a matrix $\hat{J}(y)$ by least squares:

$$\hat{J}(y)^T = \arg \min_{A \in \mathbb{C}^{n \times n}} \|hXA - \Delta F\|_F^2,$$

where X has rows ξ_i^T .

- (vi) Compute the inverse-jet correction

$$\delta = \delta_1 + \delta_2 + \delta_3,$$

where

$$\hat{J}\delta_1 = -F(y),$$

δ_2 is obtained from a symmetric second finite difference along δ_1 , and δ_3 from a mixed second/cubic stencil.

- (vii) Perform batched line search along δ and accept only validated residual decrease.

There is no Halley layer, no anchored finite-slope matrix $Q_F(a, b)$, and no Newton fallback.

2 Polynomial cost

Proposition 2.1 (Per-trial arithmetic cost). *Assume the number of atlas cells, probes, start-balancing candidates, hypercube samples M , line-search points, and local epochs are all bounded by $\text{poly}(n, d, N)$. Then one 306 trial costs $\text{poly}(n, d, N)$ arithmetic operations in the dense representation.*

Proof. A dense evaluation of all n equations over N monomials costs polynomially many operations, e.g. $\mathcal{O}(ndN)$ under direct power bookkeeping. Atlas testing and start-balancing use polynomially many such evaluations. The hypercube least-squares stage uses M evaluations, construction of an $M \times n$ design matrix, formation/solution of a least-squares problem of dimension n , and a bounded number of additional tensor-stencil evaluations. These cost $\mathcal{O}(MndN + Mn^2 + n^3)$ plus lower-order terms. Line search adds polynomially many evaluations. Hence the full trial is polynomial under polynomial budgets. $\square$

3 Random-matrix conditioning of the hypercube design

Let $X \in \mathbb{R}^{M \times n}$ have independent rows uniformly distributed in $\{\pm 1\}^n$. Then $\mathbb{E}X^T X / M = I_n$.

Lemma 3.1 (Hypercube sample covariance). *There are universal constants $c, C > 0$ such that, for every $0 < \eta < 1$ and $0 < \rho < 1$, if*

$$M \geq C\rho^{-2}(n + \log(2/\eta)),$$

then

$$\mathbb{P}\left(\left\|\frac{1}{M}X^TX - I_n\right\| \leq \rho\right) \geq 1 - \eta.$$

Consequently all singular values of $X/\sqrt{M}$ lie in $[\sqrt{1-\rho}, \sqrt{1+\rho}]$.

Proof. The rows are isotropic subgaussian vectors with bounded subgaussian norm. The stated estimate is the standard covariance concentration theorem for isotropic subgaussian designs, applied with absolute constants. Equivalently, it follows from matrix Bernstein plus a net argument on the unit sphere. $\square$

Definition 3.2 (Hypercube least-squares Jacobian). *At a point y , scale $h > 0$, and design X , set*

$$\Delta F_i = F(y + h\xi_i) - F(y), \quad i = 1, \dots, M.$$

The 306 local matrix $\hat{J}(y)$ is the least-squares map satisfying

$$hX\hat{J}(y)^T \approx \Delta F.$$

Lemma 3.3 (Taylor perturbation of the LS Jacobian). *Suppose F is analytic in $B(y, h\sqrt{n})$ and*

$$\sup_{z \in B(y, h\sqrt{n})} \|D^2F(z)\| \leq L_2.$$

On the event of [Lemma 3.1](#),

$$\|\hat{J}(y) - J_F(y)\| \leq C_1 \frac{L_2 h n}{\sqrt{1-\rho}},$$

for an absolute convention-dependent constant C_1 .

Proof. Taylor expansion gives, rowwise,

$$F(y + h\xi_i) - F(y) = hJ_F(y)\xi_i + R_i, \quad \|R_i\| \leq CL_2h^2 \|\xi_i\|^2 = CL_2h^2n.$$

Writing R for the matrix with rows R_i^T , the least-squares error satisfies

$$(\hat{J} - J_F)^T = (hX)^\dagger R.$$

On [Lemma 3.1](#), $\|(hX)^\dagger\| \leq (h\sqrt{M(1-\rho)})^{-1}$. Also $\|R\| \leq \|R\|_F \leq CL_2h^2n\sqrt{M}$. Multiplication gives the stated estimate. $\square$

Corollary 3.4 (Invertibility transfer). *If $\sigma_{\min}(J_F(y)) \geq \sigma > 0$ and*

$$C_1L_2hn/\sqrt{1-\rho} \leq \sigma/2,$$

then, on the covariance event,

$$\sigma_{\min}(\hat{J}(y)) \geq \sigma/2, \quad \|\hat{J}(y)^{-1}\| \leq 2/\sigma.$$

4 Local inverse-jet contraction

Definition 4.1 (306 regular compatible basin). *A ball $B(\zeta, r)$ around a regular zero ζ is a 306-compatible basin with parameters (σ, L_2, L_3, L_4) if*

$$\sigma_{\min}(J_F(z)) \geq \sigma, \quad \|D^k F(z)\| \leq L_k \quad (k = 2, 3, 4)$$

for $z \in B(\zeta, r)$, and the finite-stencil scales used by 306 remain inside the ball.

Theorem 4.2 (Local hypercube inverse-jet convergence). *Let $B(\zeta, r)$ be a 306-compatible basin. Choose M as in Lemma 3.1 and h satisfying the invertibility transfer condition of Corollary 3.4. Then, with probability at least $1 - \eta$ over the hypercube cloud at each local epoch, the 306 correction is well-defined. Moreover, for y sufficiently close to ζ ,*

$$\|y^+ - \zeta\| \leq C \|y - \zeta\|^4 + C' h n \|y - \zeta\|^2,$$

where C, C' depend polynomially on $\sigma^{-1}, L_2, L_3, L_4$. If $h = \mathcal{O}(\|y - \zeta\|^2 / n)$, the update is quartic up to constants; if h is only polynomially small, it remains a contraction inside a sufficiently small basin.

Proof sketch. The exact inverse-jet expansion for an analytic map with invertible Jacobian gives

$$y - \zeta + \delta_1 + \delta_2 + \delta_3 = \mathcal{O}(\|y - \zeta\|^4)$$

when $J_F(y)$ and exact multilinear derivatives are used. The 306 formula replaces $J_F(y)$ by $\hat{J}(y)$ and replaces the required second and third directional terms by symmetric finite stencils. Lemma 3.3 and Corollary 3.4 bound the inverse perturbation of $\hat{J}^{-1}$, while standard finite-difference remainders controlled by L_3, L_4 bound the stencil errors. Combining these perturbations gives the displayed inequality. The line search used in 306 only improves residual monotonicity and does not weaken the local estimate. $\square$

5 Wide Lyapunov basins for the 306 solver

The local theorem above is deliberately conservative: it certifies a small inverse-jet door near a root. Numerically, 306 behaves as if its basin is much wider. The following result explains what can be proved without understanding the global fractal boundary of the true basin: if the frontend lands in a regular residual sublevel component containing one zero, the hypercube solver with polynomial backtracking is a Lyapunov descent method on that whole component.

Let

$$\Phi(y) = \frac{1}{2} \|F(y)\|^2.$$

Definition 5.1 (Regular Lyapunov component). *A connected open set $\Omega \subset \mathbb{C}^n$ is a regular Lyapunov component for a zero ζ if:*

- (i) $\zeta \in \Omega$ and F has no other zero in Ω ;
- (ii) $\sigma_{\min}(J_F(y)) \geq \sigma > 0$ for every $y \in \Omega$;
- (iii) J_F is Lipschitz on Ω with constant L_J and $\|J_F(y)\| \leq B_J$;
- (iv) the boundary has a residual barrier: for some $0 < \alpha < \beta$,

$$\Omega_\alpha := \{y \in \Omega : \|F(y)\| \leq \alpha\} \quad \text{contains } \zeta, \quad \inf_{y \in \partial\Omega} \|F(y)\| \geq \beta.$$

Lemma 5.2 (Hypercube Gauss–Newton descent angle). *Let $y \in \Omega$ and suppose the hypercube least-squares matrix satisfies*

$$\|\hat{J}(y) - J_F(y)\| \leq \varepsilon_J \leq \sigma/4.$$

Set $p = -\hat{J}(y)^{-1}F(y)$. Then

$$D\Phi(y)[p] \leq -c_0 \|F(y)\|^2, \quad c_0 = \frac{1}{2},$$

up to replacing $1/2$ by another absolute constant under the real identification $\mathbb{C}^n \simeq \mathbb{R}^{2n}$.

Proof. Write $J = J_F(y)$ and $E = \hat{J} - J$. Since $\hat{J}p = -F(y)$,

$$Jp = \hat{J}p - Ep = -F(y) - Ep.$$

Also $\|p\| \leq \|\hat{J}^{-1}\| \|F(y)\| \leq (4/3)\sigma^{-1} \|F(y)\|$. Hence

$$\|Ep\| \leq \varepsilon_J \|p\| \leq \frac{1}{3} \|F(y)\|.$$

For $\Phi = \frac{1}{2} \|F\|^2$, the directional derivative is

$$D\Phi(y)[p] = \operatorname{Re}\langle F(y), Jp \rangle = -\|F(y)\|^2 - \operatorname{Re}\langle F(y), Ep \rangle \leq -\frac{2}{3} \|F(y)\|^2.$$

The displayed $1/2$ follows by weakening constants. $\square$

Lemma 5.3 (Polynomial Armijo step). *Under the assumptions of Definition 5.1, and with p as in Lemma 5.2, there exists*

$$\lambda_0 \geq \frac{c\sigma^2}{B_J^2 + L_J \|F(y)\|}$$

for an absolute constant $c > 0$ such that every $0 < \lambda \leq \lambda_0$ satisfying $y + tp \in \Omega$ for $0 \leq t \leq \lambda$ gives

$$\Phi(y + \lambda p) \leq \Phi(y) - c_1 \lambda \|F(y)\|^2.$$

In particular, a dyadic line-search grid down to inverse-polynomial scale finds a descent step whenever $\sigma^{-1}, B_J, L_J, \|F(y)\|$ are polynomially bounded.

Proof. The gradient of Φ is $J_F(y)^* F(y)$. On Ω , its Lipschitz constant is bounded by a polynomial expression in $B_J, L_J, \|F(y)\|$ along the considered segment; explicitly one may use $C(B_J^2 + L_J \|F(y)\|)$. Taylor's inequality gives

$$\Phi(y + \lambda p) \leq \Phi(y) + \lambda D\Phi(y)[p] + C\lambda^2(B_J^2 + L_J \|F(y)\|) \|p\|^2.$$

Use Lemma 5.2 and $\|p\| \leq C\sigma^{-1} \|F\|$; choosing λ below a constant multiple of $\sigma^2/(B_J^2 + L_J \|F\|)$ makes the quadratic term at most half the linear decrease. $\square$

Theorem 5.4 (Wide regular basin theorem for 306). *Consider the 306 solver with a polynomial dyadic backtracking line grid and hypercube sample size chosen so that the event*

$$\|\hat{J}(y) - J_F(y)\| \leq \sigma/4$$

holds at every accepted epoch, as provided by Lemmas 3.1 and 3.3 with a union bound. If an initial point y_0 lies in a regular Lyapunov component Ω and the line-search segments remain in Ω until the residual barrier is reached, then the iterates stay in Ω , $\Phi(y_k)$ decreases monotonically, and every accumulation point is the unique zero ζ . If the component constants are polynomially bounded, the number of iterations to reach residual ε is polynomial in the input parameters and $\log(1/\varepsilon)$.

Proof. At each nonzero iterate, [Lemma 5.2](#) gives a strict descent direction for Φ . [Lemma 5.3](#) gives a polynomially small line-search step with quantitative decrease. Since Φ decreases and the boundary residual is separated by β , an iterate starting in the lower part of the component cannot cross $\partial\Omega$ before reaching the root sublevel. Summing the decrease inequality gives convergence of $\Phi(y_k)$ to a limit. If the limit were positive, the uniform descent estimate on the compact sublevel region would force further decrease, contradiction. Thus $\|F(y_k)\| \rightarrow 0$. Since Ω contains only one zero, every accumulation point is ζ . Polynomial boundedness of the constants and the dyadic line grid give the iteration bound. $\square$

Corollary 5.5 (What “touching the basin” can mean rigorously). *For 306 it is enough for the frontend atlas to hit a regular Lyapunov component, not necessarily a tiny local alpha-theory ball around the zero. Therefore [Theorem 6.14](#) may be weakened/reformulated as a lower bound on the atlas measure of regular residual sublevel components. This is a larger target than a local inverse-jet door and matches the empirical behavior of 306.*

Remark 5.6. *[Theorem 5.4](#) is not yet the full proof of Smale-17, because it does not prove that Kostlan systems have inverse-polynomial atlas measure of such regular Lyapunov components. It does prove the important point suggested by the experiments: once the frontend lands anywhere in a well-conditioned descent component, the hypercube solver can be certified globally within that component.*

6 The basin-access bound

The preceding sections make the local part substantially cleaner than in the finite-slope Halley engines: once an atlas representative lies in a regular basin, random-matrix concentration controls the local matrix. The remaining issue is the global geometric one.

6.1 Projective inversion removes the affine location obstruction

The first geometric obstruction in [Theorem 6.14](#) is only apparent: a basin can be arbitrarily far away in an affine chart. The fixed Möbius/Riemann and homothety component of the 304/306 frontend is designed precisely to remove this affine-location dependence.

Lemma 6.1 (One-dimensional inversion bound). *Let $R > 0$ and let*

$$\iota_R(z) = \frac{R^2}{z}$$

be the inversion of the Riemann sphere away from 0. If $|z| > R$, then $|\iota_R(z)| < R$. More generally, if $B(\zeta, r) \subset \mathbb{C}$ satisfies $r < |\zeta| - R$, then

$$\iota_R(B(\zeta, r)) \subset B(0, R).$$

Moreover, on $B(\zeta, r)$ with $r \leq |\zeta|/2$, the image diameter is bounded by

$$\text{diam}(\iota_R(B(\zeta, r))) \leq \frac{4R^2}{|\zeta|^2} r.$$

Proof. The first claim is immediate from $|\iota_R(z)| = R^2/|z|$. If $z \in B(\zeta, r)$ and $r < |\zeta| - R$, then $|z| \geq |\zeta| - r > R$, hence the whole ball maps into $B(0, R)$. Finally

$$|\iota'_R(z)| = \frac{R^2}{|z|^2}.$$

If $r \leq |\zeta|/2$, then $|z| \geq |\zeta|/2$ on the ball, so $|\iota'_R(z)| \leq 4R^2/|\zeta|^2$. The diameter estimate follows from the mean-value bound along line segments. $\square$

Lemma 6.2 (Projective chart bound). *Let $z = (z_1, \dots, z_n) \in \mathbb{C}^n$ and embed it as $[1 : z_1 : \dots : z_n] \in \mathbb{CP}^n$. If $\|z\|_\infty > A > 1$, then there exists an index j such that the projective chart $X_j \neq 0$ has coordinates*

$$w_0 = \frac{1}{z_j}, \quad w_k = \frac{z_k}{z_j} \quad (k \neq j),$$

with

$$|w_0| < A^{-1}, \quad |w_k| \leq \frac{\|z\|_\infty}{|z_j|} \leq 1.$$

Thus every affine point far from the origin is represented in one of the n reciprocal projective charts inside the bounded polydisc

$$\{|w_0| < A^{-1}, |w_k| \leq 1\}.$$

Proof. Choose j with $|z_j| = \|z\|_\infty$. Then $|w_0| = 1/|z_j| < A^{-1}$ and $|w_k| = |z_k|/|z_j| \leq 1$. $\square$

Proposition 6.3 (Affine unboundedness is not the core obstruction). *Let $\mathcal{B} \subset \mathbb{C}^n$ be a local regular basin containing a Euclidean ball $B(\zeta, r)$. Fix $A > 1$. Either $\|\zeta\|_\infty \leq 2A$, so the basin already has a center in a bounded affine shell, or there exists a reciprocal projective chart of [Lemma 6.2](#) in which a sub-ball of $\mathcal{B}$ is mapped into a bounded polydisc. The chart distortion on that sub-ball is controlled by a polynomial expression in A , $\|\zeta\|_\infty$, and r^{-1} whenever the ball stays away from the chart hyperplane.*

Proof. If $\|\zeta\|_\infty \leq 2A$ there is nothing to prove. Otherwise choose j with $|\zeta_j| = \|\zeta\|_\infty$. On the sub-ball $B(\zeta, r')$ with $r' \leq \min(r, |\zeta_j|/2)$, the denominator z_j is bounded below by $|\zeta_j|/2$, so the reciprocal chart is holomorphic with derivative bounded by $O(1/|\zeta_j|)$ for w_0 and $O(1 + \|\zeta\|_\infty/|\zeta_j|)/|\zeta_j| = O(1/|\zeta_j|)$ for the normalized ratios after restricting to the chosen dominant coordinate. [Lemma 6.2](#) places the image center in a bounded polydisc, and the derivative bound controls the image of the sub-ball. This is exactly the multidimensional analogue of [Lemma 6.1](#). $\square$

Lemma 6.4 (Variable-radius inversion reduces compression). *Let $\zeta \in \mathbb{C}$ with $|\zeta| = \rho > 1$, and let $B(\zeta, r)$ be a basin door with $0 < r \leq \rho/4$. For an inversion*

$$\iota_R(z) = \frac{R^2}{z},$$

the image center and local radius scale satisfy

$$|\iota_R(\zeta)| = \frac{R^2}{\rho}, \quad r_{\text{image}} \asymp \frac{R^2}{\rho^2} r$$

up to absolute constants on $B(\zeta, r)$. In particular, choosing $R^2 \asymp \rho$ places the image center at scale $\asymp 1$ and compresses the door only by $\asymp \rho^{-1}$, rather than by ρ^{-2} as in the unit inversion.

Proof. The center identity is immediate. On $B(\zeta, r)$ with $r \leq \rho/4$, one has $|z| \in [3\rho/4, 5\rho/4]$. Since $|\iota'_R(z)| = R^2/|z|^2$, the derivative is comparable to R^2/ρ^2 throughout the door. The mean-value estimate gives the stated two-sided scale comparison. If $R^2 \asymp \rho$, then $|\iota_R(\zeta)| \asymp 1$ and the scale factor is $\asymp 1/\rho$. $\square$

Proposition 6.5 (Polynomial radius ladder). *Fix $C > 0$ and a dyadic ladder*

$$\mathcal{R}_C = \{2^k : 0 \leq k \leq \lceil C \log_2 N \rceil\}.$$

For every affine scale $1 \leq \rho \leq N^{2C}$ there exists $R \in \mathcal{R}_C$ such that

$$\frac{1}{2} \leq \frac{R^2}{\rho} \leq 4.$$

Consequently, every basin door centered at affine distance $\rho \leq N^{2C}$ has a ladder inversion whose image center lies in a fixed annulus and whose door radius is compressed by at most an absolute constant times $1/\rho$.

Proof. Choose k so that $2^k \leq \sqrt{\rho} < 2^{k+1}$. Since $\rho \leq N^{2C}$, this k lies in the ladder. Then $R = 2^k$ gives $R^2 \leq \rho < 4R^2$, i.e. $1/4 < R^2/\rho \leq 1$. Enlarging constants gives the displayed annulus. The compression statement follows from [Lemma 6.4](#). $\square$

Remark 6.6 (Why the 304/306 homothety ladder matters). *A unit inversion maps far roots into a bounded region but shrinks a basin door by roughly ρ^{-2} . A radius ladder allows the frontend to choose $R^2 \simeq \rho$, keeping the image center at scale one and improving the shrinkage to roughly ρ^{-1} . Thus the homothety/inversion ladder is not only a numerical trick: it is a mechanism for preventing avoidable quadratic loss in basin-door volume.*

Corollary 6.7 (Reduction of 5.4 to projective basin volume). *The arbitrary affine location of roots cannot by itself cause an exponential loss in the 306 frontend, provided the frontend contains the reciprocal/Mobius charts and a polynomial shell net. After passing to the appropriate chart, [Theorem 6.14](#) reduces to a lower bound on the projective volume or covering radius of regular compatible basin doors inside bounded chart polydiscs.*

Proof. [Lemma 6.2](#) and [Proposition 6.3](#) show that every far affine root has a bounded representative in one of finitely many reciprocal projective charts. A polynomial shell net covers the bounded radial coordinate to polynomial precision. Therefore the remaining question is not where the basin lies in affine space, but how much projective volume its regular compatible door occupies after charting and conditioning. That is precisely the volume component of [Theorem 6.14](#). $\square$

Definition 6.8 (Regular target region). *For a system F and root ζ , let $\mathcal{U}(F, \zeta)$ denote the subset of points y such that:*

- (i) *y belongs to a 306-compatible basin of ζ ;*
- (ii) *the residual/descent/equal-value/log-scale predicates of the universal atlas accept y or one of its fixed local representatives;*
- (iii) *the basin parameters obey polynomial bounds in (n, d, N) and in the projective condition number of (F, ζ) .*

Let $\mathcal{U}(F) = \bigcup_{\zeta} \mathcal{U}(F, \zeta)$ over regular roots.

6.2 Balanced projective nets: inversion plus stereographic density

The preceding inversion lemmas remove the affine-location problem. We now record the exact covering statement needed to turn a geometric basin-size lower bound into an atlas-hit probability.

Definition 6.9 (Projective inradius). *Let $S \subset \mathbb{CP}^n$ be measurable. Its projective inradius is*

$$\text{inrad}_{FS}(S) = \sup\{\epsilon > 0 : \exists x \in S \text{ with } B_{FS}(x, \epsilon) \subset S\},$$

where B_{FS} is the Fubini–Study ball.

Lemma 6.10 (Polynomial projective net hits polynomial doors). *For every $0 < \epsilon < 1$ there exists an ϵ -net $\mathcal{N}_{\epsilon} \subset \mathbb{CP}^n$ for the Fubini–Study metric with*

$$\#\mathcal{N}_{\epsilon} \leq \left(\frac{C}{\epsilon}\right)^{2n}$$

for an absolute constant C . Consequently, if a regular descent component Ω contains a projective ball of radius ϵ , then $\mathcal{N}_{\epsilon} \cap \Omega \neq \emptyset$.

Proof. The real dimension of $\mathbb{CP}^n$ is $2n$ and its Fubini–Study volume is finite. A maximal separated set at radius $\epsilon/2$ has disjoint balls of radius $\epsilon/4$ and covers by balls of radius $\epsilon/2$. Standard volume comparison on compact projective space gives cardinality at most $(C/\epsilon)^{2n}$. If $B_{FS}(x, \epsilon) \subset \Omega$, some net point lies within distance ϵ of x and therefore belongs to Ω after replacing constants by $1/2$ if desired. $\square$

Proposition 6.11 (Realization by the 306 frontend geometry). *A 306-style frontend containing affine cells, reciprocal/Möbius charts, and a dyadic homothety ladder realizes, up to constant distortion, a projective net of the form Lemma 6.10 on every polynomially bounded chart window. More precisely, for every $\epsilon \geq N^{-C_0}$ and every chart scale $\rho \leq N^{C_1}$, a dyadic ladder of $O(\log N)$ radii combined with an ϵ -mesh in the bounded chart coordinates produces an $O(\epsilon)$ -net after stereographic/projective projection, with total size polynomial in N for fixed or polynomially controlled n .*

Proof. By Lemma 6.2 and Proposition 6.5, every affine point at scale $\rho \leq N^{C_1}$ has a chart in which its representative lies in a bounded polydisc and the variable-radius inversion avoids the unnecessary ρ^{-2} compression. On bounded chart polydiscs the Fubini–Study metric is bi-Lipschitz equivalent to the Euclidean metric with constants polynomial in the chart radius. Hence an Euclidean mesh of spacing comparable to ϵ gives an $O(\epsilon)$ Fubini–Study mesh. The dyadic ladder contributes only $O(\log N)$ radii. The simplex/hypercube/projective cell families used by the frontend are deterministic low-discrepancy substitutes for such chart meshes; replacing them by an explicit mesh only strengthens the covering claim. Thus the idealized 306 frontend contains a polynomial projective net whenever ϵ is inverse-polynomial and the dimension parameters are in the Smale dense-input regime. $\square$

Theorem 6.12 (Conditional proof of basin access from projective inradius). *Assume that for Kostlan $F \sim \mathcal{K}(n, d)$, with probability at least $1 - 1/p(n, d, N)$, the union of regular Lyapunov components of Definition 5.1 contains at least one component Ω with*

$$\text{inrad}_{FS}(\Omega) \geq \frac{1}{p(n, d, N)}.$$

Then a polynomial-size 306 frontend atlas hits a regular Lyapunov component with probability at least $1/p'(n, d, N)$, and therefore the 306 solver succeeds in polynomial time with high probability, modulo the already proved hypercube concentration events.

Proof. Let $\epsilon = 1/(2p(n, d, N))$. By Lemma 6.10, an ϵ -projective net hits every component with inradius at least 2ϵ . By Proposition 6.11, the 306 frontend can realize such a net with polynomially many chart/radius/cell samples. Uniform cycling over this polynomial atlas therefore has hit probability at least the reciprocal of its polynomial cardinality; randomized repetition amplifies this to high probability. Once a point in the regular Lyapunov component is hit, Theorem 5.4 proves convergence of the backend solver, and Lemmas 3.1 and 3.3 supply the required hypercube matrix event with polynomial sample size. $\square$

Remark 6.13 (The final irreducible statement). *After inversion, stereographic/projective compactification, variable-radius homotheties, and the wide Lyapunov basin theorem, the remaining open statement is no longer about the solver or about roots being far away. It is exactly the following geometric-probabilistic claim: a typical Kostlan system has at least one regular descent component with inverse-polynomial Fubini–Study inradius, and preferably inverse-polynomial total atlas measure. Proving that statement would complete Theorem 6.14.*

Theorem 6.14 (Hypercube-Atlas Basin-Access Bound). *There exist polynomials p, q and a polynomial-size atlas window $\mathcal{A}_{n,d}^{(q)} \subset \mathcal{A}_n$ such that for Kostlan $F \sim \mathcal{K}(n, d)$,*

$$\mathbb{P}_F \left[\frac{\#\{a \in \mathcal{A}_{n,d}^{(q)} : a \in \mathcal{U}(F)\}}{\#\mathcal{A}_{n,d}^{(q)}} \geq \frac{1}{p(n, d, N)} \right] \geq 1 - \frac{1}{p(n, d, N)}.$$

Equivalently, a uniformly cycled/randomized polynomial atlas trial enters a regular compatible basin with probability at least $1/p(n, d, N)$ with high probability over F .

Remark 6.15 (Status of Theorem 5.4). *Theorem 6.14* is the desired quantitative basin-access bound. It is the exact theorem needed to close Smale-17 for 306. The present paper does not hide this point: the random-matrix stability is proved above, while *Theorem 6.14* is the global geometric measure estimate that still requires a full proof, likely combining Kostlan invariance, condition-number tail bounds, tube-volume estimates for approximate-zero basins, and the covering properties of the fixed simplex/hypercube/projective/Mobius atlas.

Proposition 6.16 (Why 306 makes the bound cleaner). *Compared with 301/304, 306 separates the two hard events:*

atlas hits a regular basin and local matrix is well conditioned.

The second event follows from *Lemmas 3.1* and *3.3* and *Corollary 3.4*. Therefore *Theorem 6.14* only needs to lower-bound regular-basin mass for the fixed atlas; it no longer needs to analyze endpoint finite-slope matrix degeneracy or Halley probe selection.

7 Complexity consequence

Theorem 7.1 (Conditional Smale-17 theorem for 306). *Assume Theorem 6.14. Choose polynomial budgets for the atlas window, hypercube samples*

$$M = C\rho^{-2}(n + \log(2T/\eta)),$$

local epochs, line-search points, and repeated trials $T = p(n, d, N) \log(1/\eta)$. Then 306 returns an approximate zero of a Kostlan square system with probability at least $1 - \eta - 1/p(n, d, N)$ in time polynomial in $(n, d, N, \log(1/\eta))$ on the high-probability regular-input event. Consequently the expected arithmetic time is polynomial provided the standard Kostlan condition-number tails are used to truncate the exceptional set.

Proof. By *Proposition 2.1*, each trial costs polynomial time. By *Theorem 6.14*, each trial enters a regular compatible basin with probability at least inverse-polynomial on the regular-input event. Repeating $T = p \log(1/\eta)$ times misses all such basins with probability at most η . Conditional on entry, *Lemma 3.1*, *Corollary 3.4*, and *Theorem 4.2* show that the hypercube inverse-jet corrector is well-defined and contracting with failure probability at most the allocated per-epoch random-matrix error; a union bound over polynomially many epochs gives total local failure at most η . Validation by residual decrease and approximate-zero certification rejects false positives. The expected-time statement follows by the usual split between the high-probability condition-number event and its polynomially integrable tail. $\square$

Corollary 7.2 (What remains to solve Smale 17). *A proof of Theorem 6.14 with polynomial p, q completes the 306 Pandrosion solution of the Kostlan average-case Smale-17 problem.*

8 Numerical evidence: complete 306 benchmark

The standalone NumPy implementation

`flow/306_pandrosion_universal_atlas_hypercube_inversejet_numpy_engine.py`

uses the 304 universal atlas/front-end and the 305 hypercube tensor inverse-jet corrector, with no Halley, no anchored finite-slope matrix, and no Newton fallback. A complete small-stress benchmark used cases

$$(2, 2), (2, 3), (2, 4), (2, 5), (3, 2), (3, 3), (3, 4), (4, 2),$$

seeds 0, 1, 2, 3, 4, count $\min(d^n, 12)$, pool 2048, epochs 28, and acceptance residual 10^{-8} . It succeeded on all 40/40 runs and recovered 425/425 requested roots.

case	successes	roots	avg trials	avg evals
(2, 2)	5/5	20/20	8.8	4400.6
(2, 3)	5/5	45/45	116.2	53338.8
(2, 4)	5/5	60/60	41.2	22154.2
(2, 5)	5/5	60/60	58.6	31736.2
(3, 2)	5/5	40/40	49.8	22749.0
(3, 3)	5/5	60/60	26.0	14154.4
(3, 4)	5/5	60/60	30.2	17195.2
(4, 2)	5/5	60/60	84.4	38948.8

These experiments do not prove [Theorem 6.14](#), but they support the separation advocated above: the fixed atlas supplies useful starts, and the hypercube local matrix avoids the finite-slope degeneracy obstruction.

9 Conclusion

306 changes the mathematical shape of Pandrosion. The local solver is no longer a hard-to-bound probe-dependent Halley/finite-slope matrix. It is a random hypercube least-squares matrix, so local non-degeneracy is governed by concentration and Taylor perturbation. Therefore the Smale-17 obstruction is now sharply localized: prove the fixed universal atlas regular-basin access bound, [Theorem 6.14](#). Once that inverse-polynomial lower bound is established, the remaining complexity proof is polynomial and standard.